

◆ 1-Mark Questions (Very Short Answers)

1. **What is RAID?**

→ RAID (Redundant Array of Independent Disks) is a data storage technology that combines multiple physical disks into a single logical unit to improve performance and provide fault tolerance.

2. **What does RAID stand for?**

→ Redundant Array of Independent Disks.

3. **Name any two RAID levels.**

→ RAID 0 (Striping) and RAID 1 (Mirroring).

4. **What is file organization?**

→ It refers to the way records are stored and arranged in a file on a storage device.

5. **What is a record?**

→ A record is a collection of related data items (fields) that represent an entity.

6. **What is indexing in a database?**

→ Indexing is a data structure technique that improves the speed of data retrieval operations on a database table.

7. **What is a B+ Tree?**

→ A B+ Tree is a balanced tree data structure used to store sorted data and allow efficient insertion, deletion, and search operations.

8. **What is hashing?**

→ Hashing is a method of mapping data of arbitrary size to fixed-size values (hash values) using a hash function.

9. **Define static hashing.**

→ Static hashing uses a fixed number of buckets, and the structure does not change even when the database grows.

10. **Define dynamic hashing.**

→ Dynamic hashing allows the hash table to grow or shrink dynamically as the number of records changes.

11. **What is query processing?**

→ Query processing is the sequence of steps used to extract data from a database efficiently.

12. What is query optimization?

→ Query optimization is the process of selecting the most efficient way to execute a given query.

◆ 2-Mark Questions

1. Write two advantages of RAID.

- Provides data redundancy (fault tolerance).
- Increases read/write performance.

2. Differentiate between RAID 0 and RAID 1.

- RAID 0 → Data striping, no redundancy, higher speed.
- RAID 1 → Data mirroring, full redundancy, higher reliability.

3. List types of file organization.

- Sequential file organization
- Direct (hashed) file organization
- Indexed file organization

4. What is the difference between index and hashing?

- Index → Ordered structure (tree-based).
- Hashing → Unordered structure (hash-based).

5. Why is indexing important?

- It improves query performance and reduces the time required to search records.

6. What is the difference between B Tree and B+ Tree?

- B Tree stores keys and data in all nodes.

- B+ Tree stores data only in leaf nodes and keys in internal nodes.

7. What is the function of a hash function?

- It converts a key into a hash value (bucket address) for quick access.

8. What is meant by selection operation in query processing?

- Selection retrieves records from a table that satisfy a specific condition (e.g., `SELECT * FROM student WHERE age > 20`).

◆ 3-Mark Questions

1. Explain RAID 0, RAID 1, and RAID 5.

- **RAID 0 (Striping):** Data divided into blocks and distributed across disks (no redundancy).
- **RAID 1 (Mirroring):** Data duplicated across disks for fault tolerance.
- **RAID 5 (Parity):** Data and parity information distributed among all disks, allowing recovery from single disk failure.

2. Explain the need for file organization.

- Efficient storage and retrieval of data.
- Minimizes disk access time.
- Provides logical and physical organization of records.

3. Explain B+ Tree structure.

- It has internal nodes (index nodes) and leaf nodes (data nodes).
- Internal nodes store only keys for navigation.
- Leaf nodes contain all actual data records and are linked sequentially.

4. What are the advantages of using hashing?

- Constant-time access to records.
- Easy insertion and deletion.
- Useful for equality searches.

5. **Explain static vs dynamic hashing.**

- **Static Hashing:** Number of buckets fixed; overflow handled by chaining.
- **Dynamic Hashing:** Number of buckets grows or shrinks based on record count.

◆ 5-Mark Questions

1. **Explain various RAID levels with advantages.**

- **RAID 0:** Striping – improves speed but no fault tolerance.
- **RAID 1:** Mirroring – full redundancy but costly.
- **RAID 5:** Block-level striping with distributed parity; balance between performance and reliability.
- **RAID 6:** Double parity; survives two disk failures.
Advantages:
 - High data availability.
 - Improved read/write speed.
 - Fault tolerance.

2. **Describe different types of file organization.**

- **Sequential File Organization:** Records stored in sequence by key.
- **Direct (Hashed) File Organization:** Hash function determines the address.
- **Indexed File Organization:** Uses index tables to locate records quickly.
Use cases:

- Sequential → Batch processing.
- Direct → Random access.
- Indexed → High-performance search.

3. Explain the structure and properties of a B+ Tree.

- All leaves are at the same level.
- Internal nodes contain only keys (no data).
- Leaf nodes contain data pointers.
- Linked leaves allow range queries.

Properties:

- Balanced structure.
- Efficient insertion/deletion/search.
- Supports ordered traversals.

4. Explain the process of query processing.

- **Steps:**
 1. **Parsing and translation** → Convert SQL to relational algebra.
 2. **Optimization** → Find best execution plan.
 3. **Evaluation** → Execute the optimized query plan.
- **Goal:** Minimize query execution cost (I/O, CPU).

◆ 10-Mark Questions

1. Explain all RAID levels with structure, advantages, and disadvantages.
 - **RAID 0 (Striping):**

- Data divided across disks → high performance, no redundancy.
 - ☒ Fast read/write. ☒ One disk failure → data loss.
 - **RAID 1 (Mirroring):**
 - Duplicate data on two disks.
 - ☒ High reliability. ☒ Expensive (50% storage wasted).
 - **RAID 5 (Distributed Parity):**
 - Data + parity across disks.
 - ☒ Tolerates one disk failure. ☒ Write speed slower.
 - **RAID 6 (Dual Parity):**
 - Two parity blocks → survives two disk failures.
 - ☒ High reliability. ☒ Slower writes.
 - **RAID 10 (Combination of RAID 1 & 0):**
 - Striped mirrors.
 - ☒ Fast + reliable. ☒ High cost.
- Conclusion:** RAID ensures data availability and performance.

2. Discuss B Tree and B+ Tree in detail.

- **B Tree:**
 - Data stored in both internal and leaf nodes.
 - Each node has multiple keys and children.
 - Easier insertion/deletion.
- **B+ Tree:**
 - Data only in leaf nodes; internal nodes store keys.

- Leaf nodes linked sequentially → efficient range queries.
- Better for database indexing.
Differences:

Feature	B Tree	B+ Tree
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Data Storage	All nodes	Leaf nodes only
Traversal	In-order	Sequential
Range Query	Complex	Easy
Used in	File Systems	Databases

3. Explain in detail static and dynamic hashing with examples.

○ Static Hashing:

- Fixed number of buckets.
- Hash function maps key to bucket.
- Overflow handled using linked list (chaining).
Example:
 If hash function = $\text{key} \% 10$, key 25 → bucket 5.

○ Dynamic Hashing:

- Directory grows with data.
- Common methods: Extendible Hashing, Linear Hashing.
Example:
 Directory doubles when overflow occurs.
Advantages of Dynamic Hashing:

- Scalable, efficient, minimizes overflow.

4. Explain query processing steps with an example query.

○ Example Query:

`SELECT name FROM student WHERE marks > 80;`

○ Steps:

- **Parsing:** SQL → relational algebra.

- **Optimization:** Choose best execution plan (index use, join order).
- **Evaluation:** Execute using storage engine.
- **Operations:**
 - Selection, projection, joins, sorting.
Goal: Fast response and minimal cost.